MIL-L-82531(OS)
18 April 1968
Superseding
NOTS XS 457 Rev. A
31 January 1968

MILITARY SPECIFICATION

FOR

LEAD, EXPLOSIVE, MK 5 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements of one type of Explosive Lead Mk 5 Mod 0. (See 6.1 for intended use and description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein, except as noted.

SPECIFICATION

FEDERAL

QQ-A-225/5	Aluminum and Aluminum Alloy Bar, Rod, Wire, or Special Shapes; Rolled, Drawn, or Cold Finished; General Specification for

STANDARDS

MILITARY

MIL-STD-105D	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC - 1340

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DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 251339	Lead, Explosive, Mk 5 Mod 0, and all drawings listed thereon
LD 268000	Primer, Electric, Mk 137 Mod 0, and all drawings listed thereon
LD 275959	Detonator Container Mk 6 Mod 0 (Inner), and all drawings listed thereon
LD 275960	Detonator Container Mk 8 Mod 0 (Outer), and all drawings listed thereon

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications

Interstate Commerce Commission

49 CFR Part 71 to 78	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations and are available from the Superintendent of Documents, Government Printing Office, Washington D.C. 20402. Orders for the above publication should cite "the latest issue and supplements thereto".)

3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the lead shall be in accordance with the requirements specified herein and with the applicable documents listed in BuWeps LD 251339.

3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the lead shall be manufactured using the methods and procedures proposed for production. The samples shall be tested as specified in Section 4 herein, and are for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Performance and product characteristics. The lead shall meet the following performance and product characteristics.

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3.3.1 Lead output and functioning. The lead shall function and produce an indentation more than 0.010 inch deep in a steel test block when subjected to the output of Detonator Mk 59 Mod 0.

3.4 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the lead shall be classified as follows:

a. Preproduction inspection (See 4.5.)

b. Quality conformance inspection (See 4.6.)

4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot", that is a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order, the number of product in each lot shall be ten thousand (10,000).

4.3.2 Preproduction sample. A preproduction sample of fifty (50) leads, manufactured in accordance with 3.2, shall be subjected to the quality conformance inspections detailed in 4.6. Acceptance of the preproduction sample shall be judged in accordance with MIL-STD-105D, inspection Level II, with an AQL of 1.0. A dud (see 6.3.1) shall be cause for rejection of the preproduction sample. Further production of the primers by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

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4.3.3 Quality conformance inspection sampling. Samples shall be selected as specified in 4.6 and shall be used to determine the acceptability of the lead inspected as specified in 4.6.

4.4 Test equipment and test conditions.

4.4.1 Quality conformance inspection equipment. The following items of test equipment are required to perform the quality conformance inspection tests:

Equipment	Capability
a. Test block assembly	See Figure 1.
b. Dial indicator: indicator stem to be approximately 0.10 inch in diameter with conical tip of 30 degrees included angle; point of tip to have 0.025-inch spherical radius	Measure differences in depth with accuracy of plus or minus 0.0005 inch. (See Figure 2.)

4.4.2 Test conditions. Unless otherwise specified, the lead shall be subjected to the preproduction and quality conformance inspection tests under the following conditions:

a. Temperature	Room ambient (65 to 95 degrees F)
b. Altitude	Normal ground atmospheric pressure
c. Humidity (relative)	Room ambient up to 95 percent

4.5 Preproduction inspection. The preproduction samples shall be subjected to the quality conformance inspections of 4.6.

4.6 Quality conformance inspection. Samples selected using the plans and proceudres set forth in Table I of this specification shall be subjected to the tests and examinations as described in 4.6.1.

4.6.1 Test procedure. The lead shall be subjected to the tests and examinations as outlined herein. Unless otherwise specified, inability of the lead to pass the tests and examinations shall constitute a failure.

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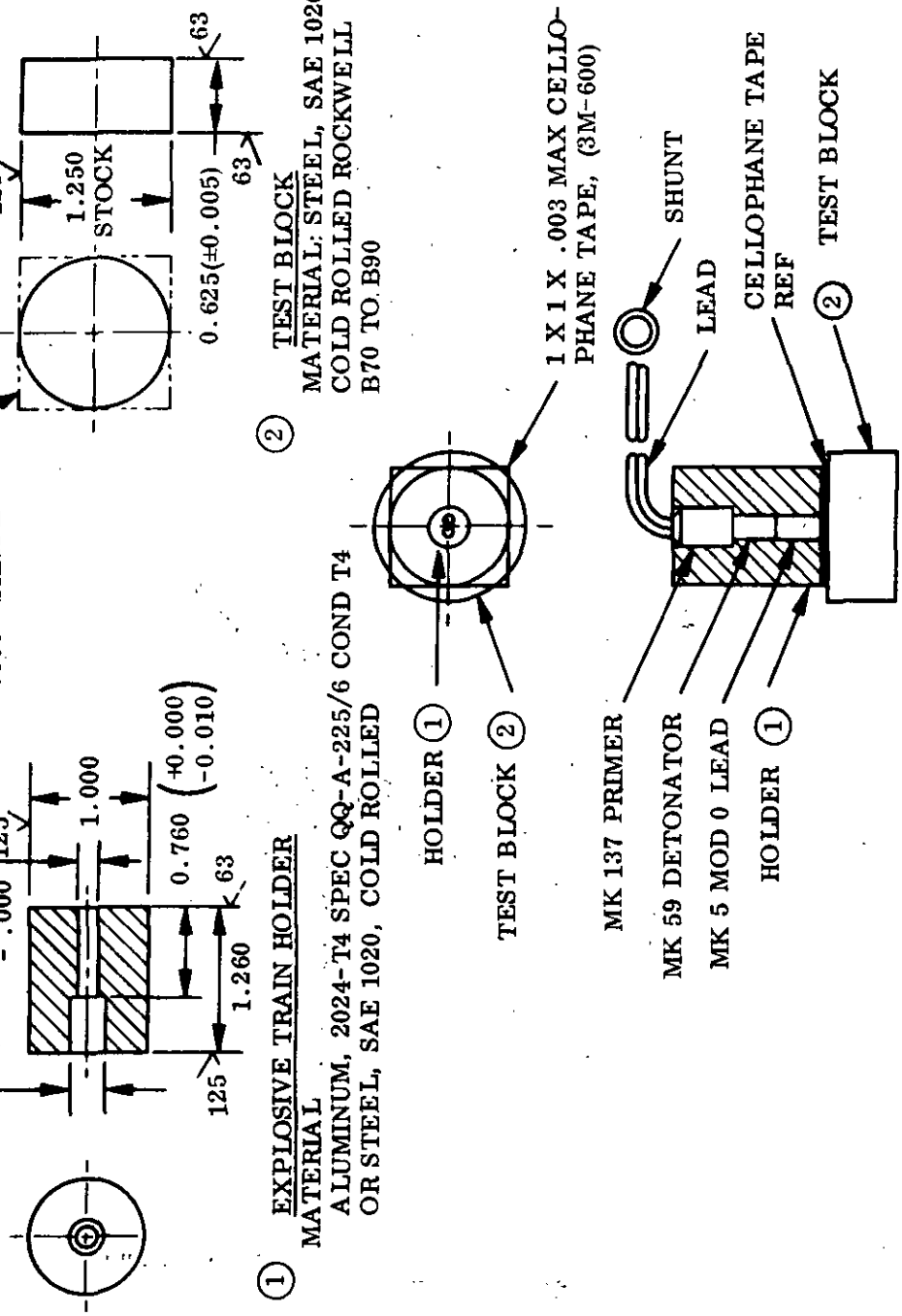


Figure 1. Explosive Train Holder and Test Block for Functioning Test

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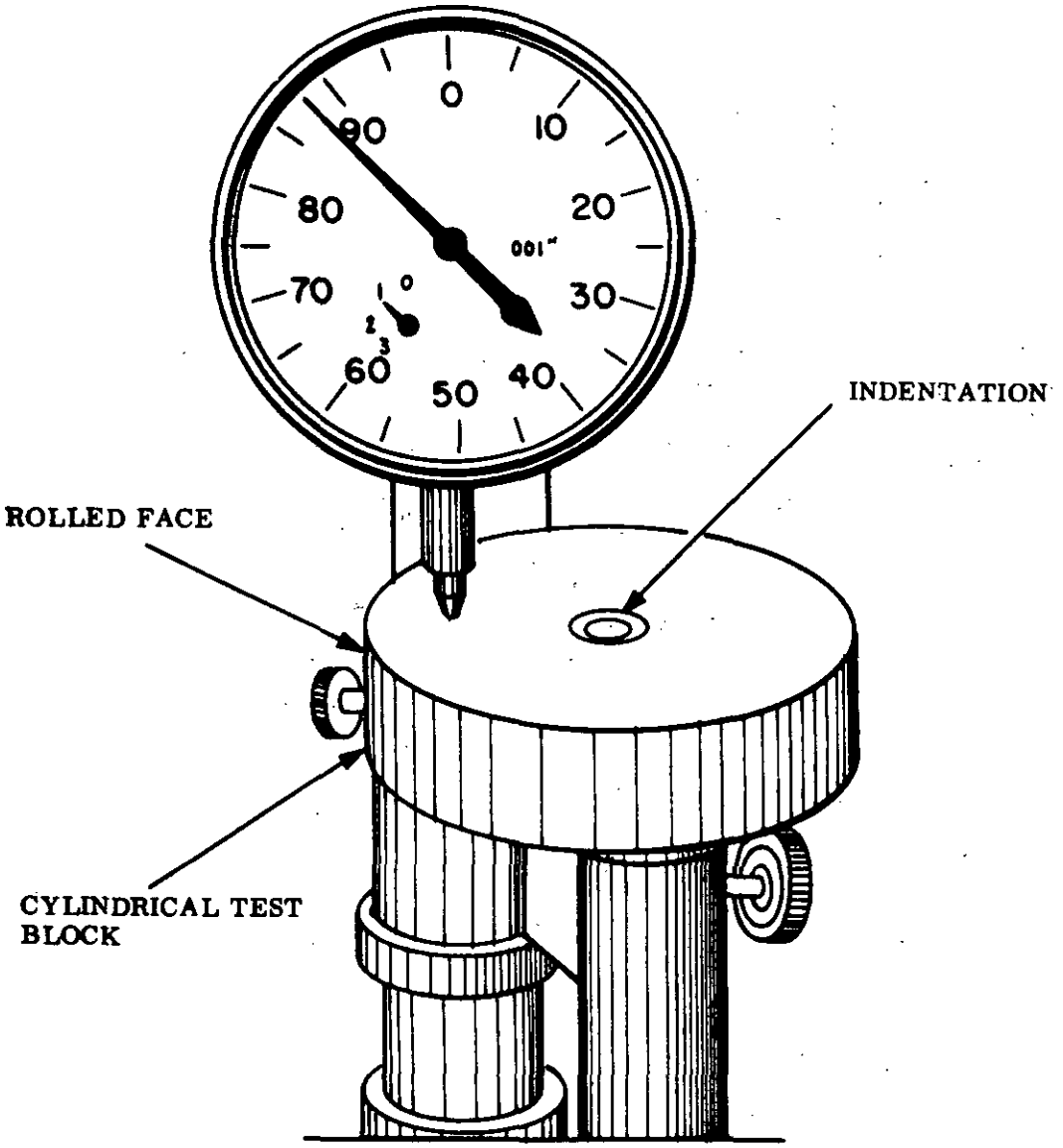


Figure 2. Correct Arrangement for Measuring Depth of Indentation After Zero Setting

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Table I. Quality Conformance Tests and Examinations and AQL

Test	Sampling Plan	AQL	Test Paragraph
Lead Output and Functioning	*	1.0	4.6.1.1
Visual Examination	**	**	4.6.1.2
Packaging, Packing, and Marking	**	**	4.7

*MIL-STD-105D Inspection Level II. The AQL of 1.0 applies to the indentation of the test block. In addition, a dud lead (see 6.3.1) shall be cause for rejection of the lot.

**One-hundred percent inspection; defect is cause for rejection of individual unit.

4.6.1.1 Lead output and functioning. This test shall be conducted using the test equipment and arrangement specified in 4.4.1a and shown in Figure 1 and 2. The lead shall be detonated by Detonator Mk 59 Mod 0, which in turn shall be initiated by either Electric Primer Mk 137 or other suitable initiator with equal output. The design of the test holder shown in Figure 1 may be modified to accommodate a suitable primer used for reliable initiation of the detonator.

4.6.1.1.1 Measurement procedure. The maximum depth of indentation produced in each test block shall be measured and recorded to the nearest 0.001 inch, using the dial indicator specified in 4.4.1b. The dial indicator shall be mounted over a horizontal surface plate so that the indicator stem is perpendicular to the surface plate, as shown in Figure 2. The test block shall be placed, indented side up, in a holder so that the bottom face of the test block is approximately parallel to the surface plate. The indicator tip shall be lowered to the bottom of the indentation, and the indicator dial set to zero. The test block holder shall then be moved so that the indicator tip contacts the surface of the test block approximately 1/8 inch from the periphery of the test block. The indicator reading shall be recorded to the nearest 0.001 inch. Four readings shall be taken at approximately equidistant points near the periphery of the indentation. The arithmetic average of the four readings shall be taken as the average depth of indentation for each test block. The average depth of indentation shall be not less than 0.010 inch.

4.6.1.2 Visual examination. Visual examination shall be made on each lead submitted for delivery to the procuring agency. The lead shall be examined for conformance to the requirements of good workmanship in 3.4 Nonconformance is cause for rejection of individual unit.

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4.7 Packaging, packing, and marking. The lead shall be examined for conformance to the prescribed packaging, packing, and marking requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. One thousand (1000) leads shall be packaged in accordance with instructions contained in the applicable documents listed in LD 275959.

5.2 Packing. Four thousand (4000) leads, packaged as specified in 5.1, shall be packed in accordance with instructions contained in the applicable documents listed in LD 275960.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, 49 CFR Part 71 to 78, and LD 275960.

5.4 Lead data cards. Five (5) cards, 3 inches by 5 inches, made from commercial white cardboard, printed as shown in Figure 3, shall be filled out for each lot of leads, and shall be distributed as follows:

a. Two cards shall accompany the shipping ticket.

b. Two cards shall be forwarded to the procuring activity.

c. One card shall be placed in each packing container.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The lead is intended for use in the motor separation and airframe separation explosive blocks of a rocket-propelled missile.

6.1.2 Description. The lead is designed to be reliably initiated by Detonator Mk 59 Mod 0 and to yield a highly brisant output. The lead is approximately 0.161 inch in diameter by 0.380 inch long, and contains approximately 120 milligrams of RDX primary explosive.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Required level of packaging

c. Required level of packing

d. Provision for supplying data cards as specified in 5.3.2

e. Any exceptions to this specification

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LEAD, EXPLOSIVE, MK 5 MOD 0 DATA CARD

Spec_____

Loaded by_____

Date: Mo._____ Yr.____

Contract No. _____

No. in Box_____

Prod. Lot Size _____

Lot Type:
( ) Regular
( ) Mixed
( ) Special

MATERIAL	MANUFACTURER	LOT OR BATCH NO.	DATE MO. YR.
Lead Cup			
Closures			
RDX Charge			

Remarks

Figure 3. Lead Data Card Format

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6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

6.3 Definitions. For the purpose of this specification the following definition has been established.

6.3.1 Dud. Any lead that fails to explode when subjected to the output of an adjacent Detonator Mk 59 Mod 0, shall be termed a dud. Failure of any lead to explode due to faulty detonator functioning or test equipment shall not be construed as a dud.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of packaging, packing, and marking shall be as specified in Section 5.

6.5 Safety precautions.

6.5.1 Special handling instructions.

6.5.1.1 Extreme care and precautions shall be exercised at all times in the handling and loading of the RDX material.

6.5.1.2 No more than 2 grams of RDX shall be present at each loading station and only when an operator is present.

6.5.2 General safety precautions. The loading assembly, and handling of explosives, ammunition, ammunition components, fuzes, detonators, primers, tracers, boosters, pyrotechnics, etc., and subassemblies thereof, involve hazardous operations, and therefore require suitable explosives safety precautions. Use of this specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. The following paragraphs list the minimum provisions that a contractor or manufacturer (loader) of these items should observe in order to fulfill his responsibility for safety. At Naval Ordnance Systems Command and other Government plants, these provisions are mandatory. Such other warnings and precautions pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail provisions of the specification.

6.5.2.1 All loading operations should be conducted in a neat and orderly manner.

6.5.2.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosive operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, and crimping.

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6.5.2.3 Personnel handling detonators, primers, delay elements, leadins, boosters, and related parts that affect functioning, should, insofar as practicable, avoid using bare fingers or improper equipment to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.2.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.2.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, kept in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intra-plant distances when outside of loading rooms. For Navy explosive loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives, shall apply.

6.5.2.6 Proper care should be exercised at all times to protect personnel from accidents, fires, or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed.

6.5.2.6.1 Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

6.5.2.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.2.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.5.2.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches, and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics, such as silk and nylon, which promote static generation, should be avoided. Additional grounding devices, such as grounded bracelets for workers, should be employed where operations are conducted with items that are usually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy-type electric primers, detonators, and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups, wrapped in bare noninsulated aluminum foil or other uncoated metal foil. During assembly and processing operations such

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sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding of these items from induced electric currents generated by sources such as lightning, static radiations from communications apparatus, radar, or high frequency heating apparatus. Where necessary for safety, humidity (see 6.5.2.4) of work rooms should be increased as required to lessen electrostatic effects without excessive moisture absorption

6.5.2.6.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heater switches, wiring, motors, lights, and test instruments, by suitable insulation, grounding, separation, or shielding. Such electric sources may initiate explosives by heat, sparks, or arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electronic soldering iron through a grounded contact. All electric-type primers, detonators, or squibs provided with wire leads, should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips of other devices should be employed during manufacturing operations involving electric primers, detonators, or squibs.

6.5.2.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and associated industrial hazards.

6.5.2.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper MATCH and NO SMOKING rules where necessary.

6.5.2.6.8 Enforce good housekeeping and maintain effective policing, inspection, and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accummulation of explosives or explosive dust and other contamination upon, and assure its removal from, floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded. Clean up any spilled material immediately.

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Naval Ordnance Systems Command
Washington, D.C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-L-82531(OS) LEAD, EXPLOSIVE, MK 5 MOD 0

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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